

WEEKLY LITERATURE REVIEW REPORT

Project: Post-Disaster Assessment using ML and Semantic Communication

Timeline: Week 3

1. KEY PAPER SUMMARY

- Paper Title: RescueNet: A High Resolution UAV Semantic Segmentation Dataset for Natural Disaster Damage Assessment
- Core Methodology Studied: The authors introduced RescueNet, a high-resolution post-disaster dataset collected via Unmanned Aerial Vehicles (UAVs) after Hurricane Michael. It provides pixel-level annotations across multiple complex structural classes (buildings, roads, pools, trees) rather than isolating a singular category. They also benchmarked state-of-the-art deep convolutional networks to demonstrate the dataset's efficacy in comprehensive scene understanding.
- Relevance to Our Project: RescueNet provides an indispensable operational framework for our custom fieldwork dataset collected near Kandy. Seeing how a professional UAV dataset categorizes complex classes like "roads" and "buildings" helps us refine our own labeling patterns. Furthermore, the immense computational complexity required to process dense pixel-level masks instead of macro classification tags reinforces our decision to implement a compact, edge-based image classification pipeline to minimize semantic communication transmission latencies.

IN-TEXT CITATION EXAMPLE

"Unlike standard aerial datasets that focus exclusively on building structures, high-resolution UAV frameworks such as RescueNet provide multi-class pixel annotations to yield a holistic understanding of a post-disaster scene."

REFERENCES (IEEE Format)

M. Rahnemoonfar, T. Chowdhury, and R. Murphy, "RescueNet: A High Resolution UAV Semantic Segmentation Dataset for Natural Disaster Damage Assessment," *Scientific Data*, vol. 10, no. 913, pp. 1-9, Dec. 2023.

2. KEY PAPER SUMMARY

- Paper Title: xBD: A Dataset for Assessing Building Damage from Satellite Imagery
- Core Methodology Studied: The authors presented xBD, the largest building damage assessment dataset containing over 850,000 building annotations. The dataset introduces a standardized ordinal scale for labeling post-disaster structural changes (No Damage, Minor Damage, Major Damage, Destroyed) across pre- and post-event satellite imagery matrices, paired with environmental factor bounding boxes for elements like fire, water, and smoke.
- Relevance to Our Project: The xBD dataset is globally recognized as the golden standard for structural change taxonomy. It directly benefits our project by providing a validated, internationally accepted ordinal labeling scheme that we can map onto our classification goals. Understanding how massive data architectures standardize "minor vs. major structural changes" ensures that our edge-deployed machine learning model can correctly classify varying degrees of disaster severity before turning them into compact feature maps.

IN-TEXT CITATION EXAMPLE

"The establishing of large-scale open-source archives like the xBD dataset provides deep learning architectures with a standardized ordinal scale to quantify damage progression across diverse disaster domains."

REFERENCES (IEEE Format)

R. Gupta et al., "xBD: A Dataset for Assessing Building Damage from Satellite Imagery," arXiv preprint arXiv:1911.09296v1, pp. 1-11, Nov. 2019.

3. KEY PAPER SUMMARY

- Paper Title: RSVLM-QA: A Benchmark Dataset for Remote Sensing Vision Language Model-based Question Answering
- Core Methodology Studied: The authors constructed RSVLM-QA, a massive remote sensing visual question answering dataset built by integrating structural data from several famous segmentation platforms (WHU, LoveDA, INRIA, iSAID). They utilized a dual-track annotation pipeline to generate descriptive, multi-turn dialogue trees capable of testing a model's advanced reasoning, counting, and spatial relation capabilities over aerial terrain.
- Relevance to Our Project: This paper sits at the absolute forefront of modern remote sensing analysis by moving beyond simple rigid classification and connecting computer vision with textual "meanings." This provides a direct conceptual foundation for our semantic communication pipeline. It proves that a scene's intelligence can be condensed into high-level descriptive intents or semantic text patterns rather than raw pixels, illustrating how future edge systems can communicate disaster intelligence back to base stations using highly minimized linguistic data payloads.

IN-TEXT CITATION EXAMPLE

"By bridging aerial imagery data with textual inquiry pipelines, Remote Sensing Vision-Language Models (RSVLMs) enable advanced spatial reasoning and contextual question answering over complex terrain matrices."

REFERENCES (IEEE Format)

X. Zi, X. Tao, J. Xiao, J. Li, and M. Prasad, "RSVLM-QA: A Benchmark Dataset for Remote Sensing Vision Language Model-based Question Answering," arXiv preprint arXiv:2508.07918v1, pp. 1-13, Aug. 2025.